Notes: Houston, James, and Marcus (JFE, 1997)

Capital Market Frictions and the Role of Internal Capital Markets in Banking

Contents

1 Big picture

This paper studies whether banks themselves face costly external finance, even though banks are usually modeled as institutions that solve other firms' financing problems. The question matters for three reasons. First, if banks cannot costlessly raise uninsured deposits, subordinated debt, or outside equity, then bank lending depends on bank cash flow and capital. Second, bank capital requirements can have real loan-supply effects. Third, the bank holding company form may create an internal capital market that reallocates scarce capital across subsidiary banks.

The central empirical finding is that **loan growth is sensitive to internally generated capital**. At the holding-company level, lending expands more when the bank holding company generates more capital internally and when it has more surplus regulatory capital. At the subsidiary-bank level, lending is more sensitive to **holding-company** cash flow and capital than to the subsidiary's own cash flow and capital.

The paper's main economic message is that bank holding companies behave like internal capital markets. Subsidiaries do not simply lend according to their own local balance sheets. Instead, the parent appears to allocate scarce capital across subsidiaries, and loan growth at one subsidiary is negatively related to loan growth at other subsidiaries in the same organization.

2 Incremental contribution of the paper

The paper makes several incremental contributions to the banking and corporate finance literatures.

1. **Moves the financing-constraints framework into banking.** The paper applies the investment-cash flow sensitivity logic of Fazzari, Hubbard, and Petersen to bank lending, treating loan growth as the bank analog of investment.

2. **Separates bank-level from holding-company-level constraints.** Prior work often studied stand-alone bank balance sheets. This paper emphasizes that most banks belong to bank holding companies, so the relevant financing constraint may be at the parent level.

3. **Provides direct evidence of internal capital markets in banking.** The paper shows that subsidiary loan growth depends strongly on capital generated elsewhere in the same holding company.

4. **Links capital requirements to loan supply.** Loan growth is lower when capital requirements bind, and the cash-flow sensitivity of lending is higher for poorly capitalized organizations.

5. **Uses underwriting fees as a cost-of-external-finance proxy.** The paper shows that banks with higher predicted external issuance costs exhibit stronger loan-growth sensitivity to internal capital.

6. **Connects banking structure to monetary policy transmission.** The results support a necessary condition for the bank lending channel: banks cannot perfectly substitute external nondeposit funding for insured deposits or retained earnings.

3 Data and measurement

3.1 Main samples

The paper uses two related samples.

- **Bank holding company sample:** 281 publicly traded bank holding companies from Federal Reserve Y-9 data over 1982–1989. Market data come from CRSP.
- **Subsidiary-bank sample:** 2,001 subsidiary banks from Call Reports, belonging to 237 multi-bank holding companies, over 1986–1989. The sample contains 4,398 subsidiary-year observations.

The 1980s sample avoids the change to risk-based capital regulation after 1989. This keeps the regulatory environment more stable for the main analysis.

3.2 Key variables

The main dependent variable is loan growth:

$$LoanGrowth_{it} = \frac{Loans_{it} - Loans_{i,t-1}}{Loans_{i,t-1}}. \tag{1}$$

The main internal-funds variable is internally generated additions to capital, scaled by lagged loans:

$$AdditionsToCapital_{it} = \frac{NetIncome_{it} + LoanLossProvisions_{it}}{Loans_{i,t-1}}. \tag{2}$$

The paper uses this capital measure because banks are constrained by regulatory capital, not simply by cash balances. Loan-loss provisions are included because they are noncash expenses and are counted in regulatory capital, subject to limits.

Surplus capital is defined as the actual Tier II capital ratio minus the regulatory minimum. The capital requirement is treated as binding when surplus capital is less than or equal to zero:

$$Bind_{it} = 1\{SurplusCapital_{it} \leq 0\}. \tag{3}$$

Control variables include market-to-book assets, securities-to-assets, log assets, subsidiary size, and indicators for whether the bank or holding company faces a binding capital requirement.

4 Models and empirical specifications

4.1 Holding-company loan-growth regressions

The first set of regressions estimates whether holding-company lending depends on internal capital after controlling for investment opportunities and liquidity:

$$LoanGrowth_{ht} = \alpha + \beta AddCap_{h,t-1} + \gamma SurplusCap_{h,t-1} + \delta Bind_{h,t-1} + \theta Q_{h,t-1} + \lambda Securities_{h,t-1} + \phi \log(Assets_{h,t-} \tag{4}$$

Some specifications include interaction terms:

$$LoanGrowth_{ht} = \cdots + \eta \left(SurplusCap_{h,t-1} \times AddCap_{h,t-1} \right) \tag{5}$$

or

$$LoanGrowth_{ht} = \cdots + \eta \left(Bind_{h,t-1} \times AddCap_{h,t-1} \right). \tag{6}$$

A positive coefficient on internal additions to capital is interpreted as evidence that internally generated capital relaxes a financing constraint. A negative interaction with surplus capital means that cash-flow sensitivity declines as the holding company becomes less capital constrained. A positive interaction with the binding-capital dummy means that loan growth is especially sensitive to internal capital when the requirement binds.

4.2 Subsidiary-bank regressions

The subsidiary-level tests ask whether loan growth at subsidiary bank b inside holding company h responds more to subsidiary capital or to capital generated elsewhere in the organization:

$$LoanGrowth_{bht} = \alpha + \beta_1 AddCap_{bht} + \beta_2 AddCap_{h,-b,t} + \gamma_1 Cap_{bht} + \gamma_2 Cap_{ht} + Controls + u_{bht}. \tag{7}$$

Here $AddCap_{h,-b,t}$ is holding-company additions to capital net of the focal subsidiary's additions. In later tests, the paper also isolates nonbank sources of parent cash flow.

4.3 Internal capital-market test with other subsidiaries' loan growth

A more direct internal capital-market test adds loan growth of related subsidiaries:

$$LoanGrowth_{bht} = \alpha + \beta_1 AddCap_{bht} + \beta_2 AddCap_{h,-b,t} + \rho LoanGrowth_{h,-b,t} + Controls + u_{bht}. \tag{8}$$

If local loan demand is correlated across subsidiaries, one might expect $\rho > 0$. A negative coefficient suggests that subsidiaries compete for a limited pool of internal capital.

5 Identification and empirical design

The paper is not a clean quasi-experiment, but it uses a sequence of empirical contrasts designed to separate financing constraints from loan-demand effects.

1. **Cash-flow sensitivity at the holding-company level.** If external finance is costly, loan growth should be increasing in internally generated capital.

2. **Capital-requirement heterogeneity.** The cash-flow sensitivity should be larger when capital requirements bind or nearly bind.

3. **Subsidiary versus parent cash flow.** If subsidiary cash flow merely proxies for local loan demand, subsidiary loan growth should respond more to its own cash flow. If the parent allocates scarce capital, subsidiary loan growth should respond more to parent-level cash flow and capital.

4. **Other-subsidiary loan growth.** A negative relationship between loan growth at the focal subsidiary and other subsidiaries is difficult to explain with common demand shocks, but natural under internal capital allocation.

5. **External-finance-cost heterogeneity.** If underwriting fees proxy for the cost of outside capital, banks with high predicted fees should show stronger loan-growth sensitivity to internal capital.

The key remaining limitation is that cash flow may still contain information about profitable lending opportunities. The paper mitigates this concern using market-to-book controls, within-organization comparisons, and the other-subsidiary tests, but the design remains observational.

6 Main results

6.1 Holding-company loan growth depends on internally generated capital

The paper finds a strong positive relationship between holding-company loan growth and internally generated additions to capital. This relationship survives controls for market-to-book ratios, liquidity, log assets, fixed effects, and alternative estimation methods.

The estimated magnitudes are economically meaningful. In the main holding-company regressions, a dollar of internally generated capital is associated with multiple dollars of loan growth. The sensitivity is lower when surplus capital is high and stronger when the capital requirement binds.

6.2 Capital requirements matter at the parent level

Simple comparisons show that holding companies below or at the regulatory minimum grow loans more slowly than those above the minimum. At the subsidiary level, loan growth is lower when the **holding company** is capital constrained, but not when only the individual subsidiary bank is undercapitalized. This distinction is central: it indicates that the relevant capital constraint operates at the consolidated organization level.

6.3 Subsidiary lending responds more to parent resources than subsidiary resources

Subsidiary loan growth is related to both the subsidiary's own additions to capital and additions to capital produced by the rest of the holding company. However, the coefficient on capital from the rest of the organization is much larger. This means the subsidiary is not a self-contained lending unit; it is part of a larger internal capital allocation system.

6.4 Loan growth is negatively related across related subsidiaries

When the paper adds loan growth at related subsidiaries, the coefficient is negative in most specifications. This is the strongest internal-capital-market evidence. It suggests that when other subsidiaries receive more capital and expand lending, the focal subsidiary grows less, consistent with competition for scarce organization-wide capital.

6.5 External financing costs reinforce the interpretation

The appendix constructs predicted underwriting fees for common stock, preferred stock, and subordinated notes. Banks with higher predicted issuance costs display greater loan-growth sensitivity to internal capital. This is consistent with the idea that internal capital matters most where external finance is most expensive.

7 Tables and figures

7.1 Summary statistics and simple capital-constraint comparisons

Table 1

Descriptive statistics for a sample of publicly traded bank holding companies and their subsidiary banks. The holding company sample includes data for 281 holding companies drawn from the Federal Reserve Y-9 tapes for the period 1982–1989. The subsidiary bank sample includes data for 2001 different bank subsidiaries drawn from the annual call reports for the subsidiary banks for the period 1986–1989. Data for the holding company sample is displayed for the total sample period and for the period 1986–1989 to facilitate comparison with the bank subsidiary sample.

	1982–1989		1982–1989	
	Mean	Median	Mean	Median
Panel A: Holding companies				
Total assets (millions)	$7,498	$2,113	$8,396	$2,452
Loan growth[a]	0.17	0.13	0.13	0.10
Asset growth	0.15	0.11	0.12	0.09
Internal additions to capital/loans$_{t-1}$[b]	0.02	0.02	0.02	0.02
Securities/total assets	0.20	0.19	0.20	0.19
Market-to-book ratio[c]	1.00	1.00	1.02	1.00
Book capital in excess of requirement/total assets[d]	0.02	0.02	0.02	0.02
Number of subsidiaries	–	–	20	15
Panel B: Bank subsidiary sample				
Total assets of bank (millions)			$293	$111
Loan growth of bank			0.08	0.10
(Internal additions to capital/loans)$_{bank}$			0.02	0.02
(Internal additions to capital/loans)$_{H\text{-net}}$[e]			0.02	0.01
(Internal additions to capital/loans)$_{Non\text{-}bank}$[f]			0.003	0.002
Securities/total assets			0.21	0.22
Lead bank total assets/total holding company total assets			0.36	0.31
((Book capital in excess of requirement)/total assets)$_{Bank}$			0.02	0.02
(Book capital in excess of requirement/total assets)$_H$			0.02	0.02

[a]Loan growth equals change in total loans outstanding divided by loans outstanding at time $t-1$.
[b]Internal additions to capital equals net income plus changes in loan loss provisions (up to the regulatory maximum).
[c]Market-to-book value of assets equals (total assets − book equity + market value equity)/total assets. Market value of equity equals market value of common stock and is taken from the Center for Research in Security Prices. The market-to-book ratio is calculated at year end for the prior year.
[d]Book capital in excess of the minimum requirement equals the bank's book capital for regulatory minimum Tier II capital ratio. Tier II capital equals common stock, preferred stock plus eligible subordinate debt and loan loss reserves. For the period 1981–1984, the minimum requirement is a 5.5%. For the period 1988–1989, the minimum requirement is 6.0%.
[e]Internal additions to capital H-net equals holding company additions to capital less the bank's additions to capital divided by total holding company loans less the loans of the subsidiary bank.
[f]Internal additions to capital Non–Bank equals holding company additions to capital, net of the aggregate additions to capital of all bank subsidiaries, divided by total holding company loans less the loans of the subsidiary bank.

Table 1. Descriptive statistics

shown in Panel A, while both groups of holding companies experienced loan growth, growth at adequately capitalized holding companies was significantly greater than growth at holding companies that failed to meet the regulatory minimum.

Table 2

Difference in loan growth based on whether minimum capital requirements are binding for a sample of bank holding companies and their subsidiary banks from 1986–1989. The holding company sample includes 281 holding companies drawn from the Federal Reserve Y-9 tapes. The subsidiary bank sample includes data for 2001 different bank subsidiaries drawn from the annual call reports for the subsidiary banks. The pooled sample for holding companies includes 954 observations during the period, and the pooled subsidiary bank sample includes 4398 observations.

	N	Loan growth	
		Mean	Median
Panel A: Holding Companies			
Capital less than or equal to regulatory minimum	31	0.041	0.047
Capital greater than regulatory minimum	923	0.132	0.107
Test statistic of difference		$t = 2.760$	$z = 3.310$
Panel B: Subsidiary Bank			
Holding company capital less than regulatory minimum	130	0.053	0.038
Holding company capital greater than regulatory minimum	4,268	0.100	0.080
Test statistic of difference		$t = 2.630$	$z = 3.120$
Bank capital less than or equal to regulatory minimum	159	0.104	0.080
Bank capital greater than regulatory minimum	4,239	0.100	0.080
Test statistic of difference		$t = -0.220$	$z = -0.430$

Table 2. Loan growth by binding capital requirement

Read: Table 1 shows that the sample consists of relatively large publicly traded bank holding companies and many small subsidiary banks. The typical holding company has internally generated additions to capital of about 2% of loans and a capital surplus of about 2% of assets. Table 2 shows that loan growth is lower when the *holding company* is at or below its capital requirement.

Economic message: The raw data already point to a consolidated capital constraint. Subsidiary loan growth is lower when the parent is constrained, but individual bank undercapitalization by itself does not reduce subsidiary loan growth in the same simple comparison.

7.2 Holding-company cash-flow sensitivity

Table 3

Loan growth and internally generated additional to capital: 1982–1989

Fixed-effects regressions and ordinary least squares (OLS) regressions relating loan growth to internal additions to capital, capital requirements, and firm financial characteristics. The sample consists of publicly traded bank holding companies over the period 1982–1989. Balance sheet inforamtion is from the Federal Reserve Y-9 tapes. The dependent variable for all regressions is calculated by subtracting loans in the previous period from loans in the current period, then dividing the total by loans in the previous period ((Loans$_t$ − loans$_{t-1}$)/loans$_{t-1}$). Fixed effects models include holding company and year effects. OLS models include dummy variabales for years (not reported). t-statistics are in parentheses.

Coefficient	Fixed effects			Ordinary least squares	
	(1)	(2)	(3)	(4)	(5)
Additions to capital/loans$_{t-1}$[a]	4.53	3.55	6.94	9.25	6.28
	(13.96)	(13.39)	(6.19)	(11.23)	(14.10)
Surplus capital/total assets$_{t-1}$[b]	0.708			0.47	
	(2.62)			(2.72)	
Bind[c]		− 0.026	− 0.20		− 0.017
		(− 2.750)	(− 2.96)		(− 2.980)
Surplus capital*(additions to capital/loans$_{t-1}$)	− 34.95			89.91	
	(− 2.92)			(− 3.63)	
Bind*(additions to capital/loans$_{t-1}$)		1.49	1.12		1.30
		(3.14)	(2.67)		(3.60)
(Market-to-book ratio)$_{t-1}$[d]	0.262	0.263	0.277	0.027	0.025
	(3.640)	(3.660)	(3.760)	(0.671)	(0.621)
(Securities-to-assets ratio)$_{t-1}$[e]	0.226	0.242	0.300	0.058	0.046
	(3.570)	(3.900)	(4.250)	(2.000)	(1.590)
Secrities/total assets$_{t-1}$*(additions to capital/loans$_{t-1}$			− 4.26		
			(− 1.56)		
log(total assets)	− 0.018	− 0.019	− 0.020	− 0.008	− 0.004
	(− 2.760)	(− 2.740)	(− 2.970)	(− 0.471)	(− 0.799)
R^2	0.240	0.245	0.246	0.357	0.353
N (categories)	1366(278)	1373(278)	1373(278)	1366(278)	1373(278)

[a]Additions to capital equal net income (before extraordinary items) plus additions to loan loss reserves.
[b]Surplus capital equals actual capital less capital required to meet minimum capital requirements.
[c]Bind = 1 if surplus capital is less than or equal to zero.
[d]Market-to-book ratio equals total assets minus book equity plus market value of common stock, divided by book value of assets.
[e]Securities-to-assets ratio equals cash, marketable securities and Fed Funds sold divided by toal assets.

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Table 3. Holding-company loan growth and internally generated capital

Table 4

Loan growth and internally generated additions to capital: 1986–1989

Between-effects regressions and ordinary least-squares (OLS) regressions relating loan growth to internal additions to capital, capital requirements, and firm financial characteristics. The sample consists of publicly traded bank holding companies over the period 1986–1989. Balance sheet information is from the Federal Reserve Y-9 tapes. The dependent variable for all regressions is calculated by subtracting loans in the previous period from loans in the current period, then dividing the total by loans in the previous period ((loans$_t$ − loans$_{t-1}$)/loans$_{t-1}$). t-statistics are in parentheses.

Coefficients	Between effects (1)	OLS (2)	Between effects (3)[a]	OLS (4)
Additions to capital/loans$_{t-1}$[b]	3.079	4.734	1.70	3.624
	(7.007)	(11.874)	(5.051)	(11.058)
Surplus capital/total assets$_{t-1}$[c]	1.762	0.506		
	(4.172)	(2.187)		
Bind[d]			− 0.058	− 0.018
			(− 4.03)	(− 2.610)
Surplus capital*(additions to capital/loans$_{t-1}$)	− 52.94	− 34.051		
	(− 3.093)	(− 2.230)		
Bind*(additions to capital/loans$_{t-1}$)			2.340	1.170
			(3.814)	(2.629)
(Market/book)$_{t-1}$[e]	0.260	0.036	0.285	0.0274
	(2.074)	(6.690)	(2.288)	(0.510)
(Securities/total assets)$_{t-1}$[f]	0.442	− 0.001	0.458	− 0.013
	(4.34)	(− 0.035)	(4.553)	(− 0.352)
log (total assets)	− 0.145	− 0.003	− 0.137	− 0.004
	(− 6.628)	(− 1.583)	(− 6.294)	(− 1.593)
R^2	0.262	0.311	0.266	0.345
N (categories)	765(274)	765	775(274)	775

[a]Results for fixed effects techniques are similar to those reported.
[b]Additions to capital equals net income before extraordinary items plus additions to loan loss reserves.
[c]Surplus capital equals actual capital less capital required to meet minimum capital requirements. Capital is Tier II. Common stock, preferred stock plus eligible subordinate debt and loan loss reserves, also known as Tier II capital.
[d]Bind equals one if surplus capital is less than or equal to zero.
[e]Market-to-book ratio equals total assets minus book equity plus market value of common stock, divided by book value of assets.
[f]Securities-to-assets ratio equals cash, marketable securities and Fed Funds sold, divided by total assets.

Table 4. Holding-company loan growth, 1986–1989

Read: Table 3 reports the main holding-company loan-growth regressions for 1982–1989. Internally generated additions to capital have large positive coefficients. Surplus capital interactions are negative, and binding-capital interactions are positive. Table 4 repeats the analysis for the shorter 1986–1989 period and produces similar conclusions.

Economic message: Loan growth is not independent of bank internal funds. The effect is strongest precisely where theory predicts it should be strongest: among organizations with little surplus capital or binding regulatory requirements.

7.3 External financing costs and loan-growth sensitivity

Table 5
Underwriting fees and investment-cashflow senstivities

Fixed effects regressions relating loan growth to internal additions to capital, expected underwriting fees, and firm financial characteristics by type of security issued. The sample consists of publicly traded bank holding companies over the period 1982–1990. Balance sheet information is from the Federal Reserve Y-9 tapes. The dependent variable for all regressions is calculated by subtracting loans in the previous period from loans in the current period, then dividing the total by loans in the previous period $((\text{Loans}_t - \text{loans}_{t-1})/\text{loans}_{t-1})$. t-statistics are in parentheses.

Variable	Predicted fees for common stock		Predicted fees for preferred stock		Predicted fees for subordinated notes	
	(1)	(2)	(3)	(4)	(5)	(6)
Additions to capital/loans$_{t-1}$	4.336	3.172	4.258	3.100	4.354	3.185
	(13.34)	(12.32)	(13.09)	(12.02)	(13.39)	(12.37)
High Fees$_{t-1}$*(additions to capital/loans$_{t-1}$[b])	0.659	0.709	0.755	0.809	0.589	0.639
	(3.545)	(3.835)	(4.309)	(4.648)	(3.275)	(3.583)
Surplus capital/total assets$_{t-1}$[c]	1.082		1.044		1.077	
	(3.918)		(3.789)		(3.889)	
Surplus capital*(additions to capital/loans$_{t-1}$)	− 42.86		− 42.25		− 43.14	
	(− 3.609)		(− 3.572)		(− 3.627)	
Bind[d]		− 0.027		− 0.027		− 0.027
		(− 2.949)		(− 2.942)		(− 2.948)

Table 5. Predicted underwriting fees and cash-flow sensitivity, part 1

Bind*(additions to capital/loans$_{t-1}$)		1.922		1.917		1.918
		(4.092)		(4.095)		(4.076)
Securities-to-total assets ratio$_{t-1}$[e]	0.205	0.213	0.211	0.219	0.208	0.216
	(3.400)	(3.536)	(3.505)	(3.639)	(3.440)	(3.583)
Market-to-book ratio[f]	0.194	0.196	0.252	0.258	0.193	0.195
	(2.405)	(2.421)	(3.091)	(3.160)	(2.393)	(2.409)
log (Total assets$_{t-1}$)	− 0.019	− 0.017	− 0.020	− 0.019	− 0.019	− 0.018
	(− 2.714)	(− 2.488)	(− 2.875)	(− 2.665)	(− 2.833)	(− 2.615)
R^2	0.231	0.238	0.239	0.246	0.230	0.236
N (categories)	1285 (251)	1285 (251)	1285 (251)	1285 (251)	1285 (251)	1285 (251)

[a]Additions to capital equals net income before extraordinary items plus additions to loan loss reserves.
[b]High Fees equals one if the firms predicted fees from the Underwriting Fees model in Table 10, corrected for selectivity bias, are greater than the median predicted fees, and equals zero otherwise.
[c]Surplus capital equals actual capital less capital required to meet minimum capital requirements. Capital is defined as Tier II capital which equals common stock, preferred stock plus eligible subordinate debt and loan loss reserves.
[d]Bind equals one if surplus capital is less than or equal to zero, and equals zero otherwise.
[e]Securities-to-assets ratio equals cash, marketable securities and Fed Funds sold, divided by total assets.
[f]Market-to-book equals total assets minus book equity plus market value of common stock, divided by book value of assets.

Table 5. Predicted underwriting fees and cash-flow sensitivity, part 2

Read: Table 5 splits holding companies by predicted underwriting costs for common stock, preferred stock, and subordinated notes. The regressions ask whether loan-growth sensitivity to internal capital is stronger when outside capital is more expensive.

Economic message: The results support the financing-frictions interpretation. Banks predicted to face higher external issuance costs rely more heavily on internally generated capital when expanding lending.

7.4 Subsidiary-level internal capital allocation

Table 6
Subsidiary loan growth and holding company internal additions to capital

Between-effects regressions and ordinary least squares (OLS) regressions relating subsidiary loan growth to internal additions to capital, capital requirements, and financial characteristics of subsidiaries and bank holding companies. The sample consists of 2001 subsidiaries of 237 multiple bank holding companies from 1986–1989. Data for subsidiary banks is from the call report. Balance sheet data on holding companies are from the Federal Reserve Y-9 tapes. Market value of holding company common stock is from the CRSP master tapes. The small bank sample consists of banks with less than 15% of total holding company assets. The dependent variable for all regressions is calculated by subtracting loans in the previous period from loans in the current period, then dividing the total by loans in the previous period $((\text{Loans}_t - \text{loans}_{t-1})/\text{loans}_{t-1})$. t-statistics are in parentheses.

Variable	Overall sample				Small bank sample			
	Between effects (1)	OLS (2)	Between effects (3)	OLS (4)	Between effects (1)	OLS (2)	Between effects (3)	OLS (4)
(Additions to capital/loans$_{t-1}$)$_{\text{other}}$[a]	2.002	1.576	1.78	1.65	2.04	1.960	2.21	2.11
	(7.371)	(7.394)	(6.00)	(7.69)	(7.11)	(7.887)	(7.54)	(8.37)
(Additions to capital/loans$_{t-1}$)$_{\text{Bank}}$[b]	0.263	0.251	0.257	0.26	0.251	0.249	0.248	0.247
	(4.545)	(7.538)	(4.44)	(7.68)	(4.371)	(7.328)	(4.356)	(7.238)
(Surplus capital/total assets)$_{\text{holding company}}$[c]	− 0.100	− 0.021			− 0.034	− 0.132		
	(− 0.031)	(− 0.078)			(− 0.100)	(− 0.438)		
(Surplus capital/total assets)$_{\text{Bank}}$[d]	0.332	0.336			0.298	0.274		
	(1.545)	(0.100)			(1.397)	(2.681)		
Bind$_{\text{holding company}}$[e]			− 0.056	− 0.05			− 0.058	− 0.052
			(− 2.920)	(− 2.54)			(− 2.984)	(− 2.582)

Table 6. Subsidiary loan growth and parent capital, part 1

Bind$_{\text{Bank}}$			0.025	0.031			0.037	0.027
			(1.106)	(1.75)			(1.124)	(1.421)
Market-to-book$_{\text{holding company}}$[f]	0.555	0.578	0.528	0.559	0.660	0.690	0.631	0.665
	(4.188)	(7.011)	(4.07)	(6.737)	(4.068)	(7.007)	(3.951)	(6.697)
(Securities-to-assets)$_{\text{Bank}}$[g]	0.067	0.067	0.074	0.067	0.043	0.036	0.048	0.041
	(2.196)	(2.593)	(2.43)	(2.61)	(1.348)	(1.315)	(1.524)	(1.500)
R^2	0.056	0.066	0.056	0.065	0.06	0.07	0.06	0.07
N (categories)	4375 (2001)	4375	4375 (2001)	4375	3974 (1839)	3974	3974 (1839)	3974

[a]Additions to capital H-Net equals holding company additions to capital less the bank's additions to capital divided by total holding company loans, less the loans of the subsidiary bank.
[b]Additions to capital Bank equals subsidiary additions to capital divided by loans of the subsidiary bank.
[c]Surplus capital equals actual capital/less capital required to meet minimum capital requirements for the holding company.
[d]Surplus capital equals actual capital held by the subsidiary bank less the minimum required for the subsidiary.
[e]Bind equals one if bank capital ratio is less than or equal to the regulatory minimum.
[f]Market-to-book equals total assets minus book equity plus market value of common stock, divided by book value of assets.
[g]Securities-to-assets equals cash, marketable securities and Fed Funds sold, divided by total assets.

Table 6. Subsidiary loan growth and parent capital, part 2

Read: Table 6 is the key subsidiary-level test. Subsidiary loan growth is positively related to the subsidiary's own capital additions, but it is much more strongly related to the capital generated by other parts of the holding company. The holding-company capital constraint matters more than the bank-level capital constraint.

Economic message: The subsidiary bank is not the relevant standalone unit for financing-constraint analysis. The evidence suggests that the parent allocates capital across subsidiaries and that internal funds produced elsewhere in the organization support lending at the focal bank.

7.5 Nonbank sources and related-subsidiary loan growth

Table 7

Subsidiary loan growth and holding company internal additions to capital from nonbank sources

Between effects and ordinary least squares (OLS) regressions relating subsidiary loan growth to internal additions to capital, capital requirements and financial characteristics of subsidiaries and bank holding companies. The sample consists of 2001 subsidiaries of 237 multiple bank holding companies from 1986–1989. Data for subsidiary banks is from the call report. Balance sheet data on holding companies are from the Federal Reserve Y-9 tapes. Market value of holding company common stock is from the CRSP master tapes. The small bank sample consists of banks with less than 15% of total holding company assets. The dependent variable for all regressions is calculated by subtracting loans in the previous period from loans in the current period, then dividing the total by loans in the previous period ($(\text{Loans}_t - \text{loans}_{t-1})/\text{loans}_{t-1}$). t-statistics are in parentheses.

Variable	Overall sample		Small bank subsidiaries	
	Between effects (1)	OLS (2)	Between effects (1)	OLS (2)
$(\text{Additions to capital/loans}_{t-1})_{\text{Non-bank}}$[a]	1.006	0.844	2.504	1.696
	(4.432)	(6.347)	(6.576)	(8.662)
$(\text{Additions to capital/loans}_{t-1})_{\text{Bank}}$[b]	0.265	0.136	0.244	0.074
	(5.061)	(2.979)	(4.434)	(1.584)
$\text{Bind}_{\text{Holding company}}$[c]	− 0.035	− 0.022	− 0.0345	− 0.020
	(− 1.308)	(− 1.966)	(− 1.197)	(− 1.684)
$\text{Bind}_{\text{Bank}}$[d]	0.014	− 0.024	0.0292	− 0.0149
	(0.613)	(− 2.102)	(1.198)	(− 1.265)
$\text{Market-to-book ratio}_{\text{Holding company}}$[e]	0.994	0.490	1.024	0.528
	(9.038)	(9.962)	(7.823)	(9.094)
$(\text{Securities-to-assets ratio})_{\text{Bank}}$[f]	0.085	0.150	0.0860	0.162
	(2.705)	(9.424)	(2.533)	(9.689)
R^2	0.08	0.097	0.093	0.114
N (categories)	4398 (2001)	4398	3965 (1839)	3965

[a]Additions to capital/loans$_{t-1}$ $_{\text{Non-bank}}$ equals holding company additions to capital net of the aggregate additions to capital of all bank subsidiaries divided by total holding company loans less the loans of the subsidiary bank.

[b]Additions to capital/loans$_{t-1}$ $_{\text{Bank}}$ equals subsidiary additions to capital divided by loans of the subsidiary bank.

[c]$\text{Bind}_{\text{Holding company}}$ equals one if bank holding company capital ratio is less than the loans of the subsidiary bank.

[d]$\text{Bind}_{\text{Bank}}$ equals one if bank capital ratio is less than or equal to the regulatory minimum.

[e]Market-to-book ratio equals total assets minus book equity plus market value of common stock, divided by book value of assets.

[f]Securities-to-assets ratio equals cash, marketable securities and Fed Funds sold, divided by total assets.

Table 7. Internal additions to capital from nonbank sources

Table 8

Subsidiary loan growth, holding company internal additions to capital, and loan growth in related subsidiaries

Between effects regressions and ordinary least squares (OLS) regressions relating subsidiary loan growth to internal additions to capital, capital requirements, and financial characteristics of subsidiaries and bank holding companies. Data for subsidiary banks is from the call report. Balance sheet data on holding companies are from the Federal Reserve Y-9 tapes. Market value of holding company common stock is from the CRSP master tapes. The small bank sample consists of banks with less than 15% of total holding company assets. The dependent variable for all regressions is calculated by subtracting loans in the previous period from loans in the current period, then dividing the total by loans in the previous period ($(\text{loans}_t - \text{loans}_{t-1})/\text{loans}_{t-1}$). t-statistics are in parentheses.

Variable	Overall Sample		Small bank subsidiaries	
	Between effects (1)	OLS (2)	Between effects (1)	OLS (2)
$(\text{Additions to capital/loans}_{t-1})_{\text{Nonbank}}$[a]	0.182	1.971	3.58	2.415
	(3.040)	(12.703)	(11.09)	(13.068)
$(\text{Additions to capital/loans}_{t-1})_{\text{Bank}}$[b]	− 0.063	0.0618	0.166	0.024
	(− 2.752)	(1.385)	(2.660)	0.516
$\text{Bind}_{\text{Holding company}}$[c]	0.022	− 0.0417	− 0.067	− 0.046
	(1.160)	(− 3.715)	(− 2.765)	(− 3.0.966)
$\text{Bind}_{\text{Bank}}$[d]	− 0.138	− 0.012	0.020	− 0.007
	(− 4.884)	(− 1.101)	(1.006)	(− 0.648)
$\text{Market-to-book ratio}_{\text{Holding company}}$[e]	0.600	0.003	− 0.171	− 0.035
	(− 5.814)	(0.231)	(− 5.540)	(− 2.169)
$(\text{Securities-to-assets ratio})_{\text{Bank}}$[f]	0.0234	0.312	0.660	0.358
	(0.845)	(6.292)	(5.777)	(6.118)
R^2	0.06	0.108	0.0015	0.103
		(6.825)	(0.959)	(6.119)
N (categories)	4307 (2001)	0.143	0.06	0.150

[a]Additions to capital/loans$_{t-1}$ $_{\text{Non-bank}}$ equals holding company additions to capital net of the aggregate additions to capital of all bank subsidiaries divided by total holding company loans less the loans of the subsidiary bank.

[b]Additions to capital/loans$_{t-1}$ $_{\text{Bank}}$ equals subsidiary additions to capital divided by loans of the subsidiary bank.

[c]$\text{Bind}_{\text{Holding company}}$ equals one if bank holding company capital ratio is less than the loans of the subsidiary bank.

[d]$\text{Bind}_{\text{Bank}}$ equals one if bank capital ratio is less than or equal to the regulatory minimum.

[e]Market-to-book ratio equals total assets minus book equity plus market value of common stock, divided by book value of assets.

[f]Securities-to-assets ratio equals cash, marketable securities and Fed Funds sold, divided by total assets.

Table 8. Loan growth in related subsidiaries

Read: Table 7 shows that subsidiary loan growth is related to holding-company internal additions to capital from nonbank sources, not only from other banks. Table 8 adds loan growth in related subsidiaries and finds negative coefficients in most specifications.

Economic message: Table 7 strengthens the case that parent-level resources matter. Table 8 provides a more direct internal-capital-market signal: capital appears scarce within the organization, so expansion at other subsidiaries can crowd out expansion at the focal subsidiary.

7.6 Capital issuance and underwriting-cost model

Table 9
Summary of Security Issuances from 1982–1989 for a sample of 278 publicly traded bank holding companies. Security issuances are from the Investment Dealers Digest. Announcement dates are from the Dow Jones News Retrieval Service. Underwriter fees are from the Investment Dealers Digest and are expressed as a percent of offer size. Abnormal returns are calculated using the methodology described in Mikkelson and Partch (1986). Market model parameters are estimated from 260 trading days to 10 days prior to the security issuing announcement.

Year	Common stock	Preferred stock	Subordinated notes	Total
1982	1	10	4	15
1983	4	19	4	27
1984	9	8	17	34
1985	19	6	26	51
1986	28	4	16	48
1987	7	8	35	50
1988	1	4	9	14
1989	4	6	15	25
Total	73	65	126	264
Mean offer size (thousands)	78,304	103,825	111,675	100,515
Mean offer size relative to total assets	1.09%	0.71%	1.29%	1.09%
Mean underwriter fees	4.49%	2.69%	1.35%	2.55%
Mean abnormal return	− 1.29%	− 0.08%	− 0.04%	− 0.42%
	($z = -4.54$)	($z = -1.03$)	($z = -0.35$)	($z = -3.01$)

that employed by Calomiris and Himmelberg (1995). The results of the estimated probit model show that the probability of a stock issue is positively related to the firm's stock return over the preceding three months, positively

Table 9. Summary of bank holding company security issuances

Table 10
Regression models relating underwriting fees to firm characteristics, correcting for selectivity bias using Heckman's two step procedure. The sample consists of 278 publicly traded bank holding companies over the period 1982–1989. In the models, the variable kinds equals one if surplus capital is less than or equal to zero, and the variable is set to zero otherwise. Additions to capital equal net income, before extraordinary items, plus additions to loan loss reserves. The ratio of securities to assets is calculated using the sum of cash marketable securities and Fed funds, divided by total assets. The market-to-book ratio equals total assets minus book equity plus market value of common stock divided by book value of assets. Risk equals the standard deviation in daily stock returns calculated over 250 trading days prior to the security issue. In the probit model, the dependent variable takes a value of one to represent the event of a capital issue. The percentage fee model estimates the percentage fee paid by the issuing holding company. The number of observations for all models is 1,498. t-statistics are in parentheses. Balance sheet data are from the Federal Reserve Y-9 tapes.

Variable	Common stock models		Preferred stock models		Subordinated note models	
	Probit model	Percentage fee model	Probit model	Percentage fee model	Probit model	Percentage fee model
Bind	0.412	0.165	0.555	0.014	− 3.185	0.001
	(2.883)	(1.694)	(3.349)	(3.353)	(− 0.728)	(0.259)
log (Total assets$_{t-1}$)	0.054	− 0.008	0.312	− 0.002	0.321	− 0.007
	(1.135)	(− 6.757)	(4.793)	(− 1.087)	(7.072)	(− 3.783)
Lag (Additions to capital/loans)	15.28	− 0.375	4.422	− 0.096	13.47	− 0.384
	(2.965)	(− 2.226)	(0.701)	(− 0.533)	(2.723)	(− 2.421)
(Securities/total assets)$_{t-1}$	− 1.148	0.046	− 0.867	− 0.079	− 1.393	0.001
	(− 1.324)	(1.957)	(− 0.682)	(− 2.289)	(− 1.645)	(0.059)
Market-to-book	− 0.392	− 0.038	− 9.538	− 0.432	− 1.729	0.004
	(− 0.396)	(− 1.852)	(− 2.763)	(− 3.434)	(− 0.825)	(0.075)
Risk	− 4.999	0.453	− 0.412	0.001	0.412	0.223
	(− 0.713)	(1.504)	(− 0.073)	(0.005)	(0.090)	(1.357)
3 month avg stock price/ 36 month avg stock price	12.46		12.32		1.673	
	(2.681)		(2.212)		(0.351)	
Inverse mills ratio		− 0.014		0.021		− 0.001
		(− 2.045)		(8.825)		(− 0.132)
Chi-squared (p-value)		44.54		127.03		133.13
		(0.001)		(0.000)		(0.000)

Table 10. Underwriting fee models

Read: Table 9 documents 264 capital issuances from 1982–1989, including common stock, preferred stock, and subordinated notes. Common stock issuance has a negative mean abnormal return. Table 10 estimates predicted underwriting fees using issuer characteristics and selection controls.

Economic message: The paper does not simply assume external finance is costly. It provides issuance evidence and a predicted-fee model, then uses those predicted costs to show that more externally constrained banks have stronger loan-growth sensitivity to internal capital.

8 Short summary

Houston, James, and Marcus study whether bank holding companies face financing frictions and whether these frictions shape lending. Their answer is yes. Loan growth is sensitive to internally generated capital, especially for banks near regulatory capital limits. At the subsidiary level, loan growth depends more on holding-company capital and cash flow than on the subsidiary's own capital and cash flow. Related subsidiaries' lending growth often enters negatively, suggesting that subsidiaries compete for scarce parent-level capital.

The paper therefore provides evidence that bank holding companies operate internal capital markets. These internal capital markets matter because they mediate how capital requirements, bank profitability, and external finance costs translate into loan supply.

9 Conclusion

The core conclusion is that **bank holding companies are financially constrained organizations that allocate scarce internal capital across subsidiaries**. Banks are not merely frictionless intermediaries that channel funds to other firms. They also face adverse-selection and moral-hazard problems when raising outside capital.

This conclusion has three important implications. First, bank capital requirements can affect credit supply because constrained banks cannot costlessly replace lost capital with external funds. Second, the bank lending channel of monetary policy is more plausible when banks face costly external finance. Third, the organization of banks into holding companies matters because the parent can reallocate capital across subsidiaries through an internal capital market.

The incremental contribution of the paper is to show that capital-market frictions and internal-capital-market logic apply inside banking organizations themselves. The paper shifts the unit of analysis from the individual bank to the bank holding company and shows that consolidated capital and internal capital allocation are central to understanding bank loan growth.